Safa Cicek

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Experience

Waymo **February 2021 - Now**
Staff Perception Engineer *Mountain View, California*

- Led the design, pretraining, and finetuning of multi-modal Vision Language Models (VLMs) on GPUs and TPUs, utilizing sensor fusion for multi-task models with 2D/3D object detection, depth prediction, and segmentation tasks.
- Defined the technical roadmap and OKRs for multi-task learning, orchestrating alignment between Perception and Infrastructure orgs to maximize model throughput and training efficiency at scale.
- Primary contributor to the perception codebase, maintaining high system stability and code velocity.
- Mentored summer interns, providing technical direction to ensure successful delivery of research projects, including autoregressive probabilistic world models (VQVAE/Transformers) and sparse transformer architectures for efficient scaling.
- Spearheaded the perception stack migration from TensorFlow to JAX/Flax, establishing foundational infrastructure for mixed-precision training and model freezing.
- Leveraged data scaling laws to drive a strategic dataset expansion, resulting in substantial improvements in critical perception metrics.
- Developed an experimentation platform for task owners, standardizing hill-climbing workflows adopted by multiple downstream teams.
- Architected the automated daily training and evaluation ecosystem for the production perception model, eliminating costly manual regression testing and standardizing the release workflow for 20+ task owners and 40+ tasks.

Waymo **June 2020 - September 2020**
Intern *Mountain View, California*

- Applied graph neural networks (GNN) to object detection tasks.

Adobe Research **June 2019 - September 2019**
Intern *San Jose, California*

- Applied GAN models (e.g. style GAN) to unsupervised domain adaptation tasks.

Honda Research Institute **June 2018 - September 2018**
Intern *Mountain View, California*

- Applied multi-agent reinforcement learning algorithms to motion planning tasks.

Education

UCLA **2015-2020**
PhD in Electrical and Computer Engineering *Los Angeles, California*

- In UCLA Vision Lab, under the supervision of Professor Stefano Soatto.
- Committee: Stefano Soatto, Lieven Vandenberghe, Paulo Tabuada, Guy Van den Broeck.
- Thesis: Visual Learning with Weak Supervision.

UCLA **2015-2017**
MS in Electrical Engineering *Los Angeles, California*

Bilkent University **2011-2015**
BS in Electrical Engineering *Turkey*

- Graduated within 3.5 years as a Valedictorian. 3.97 GPA out of 4.00.

Ankara Science High School **2007-2011**
Turkey

Skills

Deep learning libraries: JAX, TensorFlow, Pytorch, Flax, Keras.
Programming languages: Python (**Google readability**), C++, SQL, Matlab, Java.

Publications

- **Safa Cicek** and Alhussein Fawzi and Stefano Soatto, Saas: Speed as a Supervisor for Semi-Supervised Learning, European Conference on Computer Vision (ECCV). 2018.
- **Safa Cicek**, Stefano Soatto, Unsupervised Domain Adaptation via Regularized Conditional Alignment, International Conference on Computer Vision (ICCV) as **oral (%4.6)**. 2019.
- Alex Wong, **Safa Cicek**, Stefano Soatto, Learning Topology from Synthetic Data for Unsupervised Depth Completion, IEEE Robotics and Automation Letters (RAL). 2021.
- Alex Wong, **Safa Cicek**, Stefano Soatto, Targeted Adversarial Perturbations for Monocular Depth Prediction, Conference on Neural Information Processing Systems (NeurIPS). 2020.
- **Safa Cicek**, Ning Xu, Zhaowen Wang, Hailin Jin, Stefano Soatto, Generative Feature Disentangling for Unsupervised Domain Adaptation. European Conference on Computer Vision Workshop (ECCVW). 2020.
- **Safa Cicek**, Ning Xu, Zhaowen Wang, Hailin Jin, Stefano Soatto, Spatial Class Distribution Shift in Unsupervised Domain Adaptation: Local Alignment Comes to Rescue. Asian Conference on Computer Vision (ACCV). 2020.
- **Safa Cicek**, Stefano Soatto, Input and Weight Space Smoothing for Semi-supervised Learning, International Conference on Computer Vision Workshop (ICCVW). 2019.
- **Safa Cicek**, Alireza Nakhaei, Stefano Soatto, MARL-PPS: Multi-agent Reinforcement Learning with Periodic Parameter Sharing, International Conference on Autonomous Agents and Multiagent Systems (AAMAS). 2019.
- Wong, Alex, et al. "Small Lesion Segmentation in Brain MRIs with Subpixel Embedding." International MICCAI Brainlesion Workshop, 2021.
- Chung, Younjoon, et al. "ETA: Energy-based Test-time Adaptation for Depth Completion." Proceedings of the IEEE/CVF International Conference on Computer Vision. 2025.

Patents

- Controlled style-content image generation based on disentangling content and style. 11,776,180, 2023.
- Contrastive learning for object detection. 11,756,309, 2023.
- System and method for multi-agent reinforcement learning with periodic parameter sharing. 11,657,251, 2023.
- Trajectory prediction from multi-sensor fusion. 18/517,750, 2025.
- Methods and Systems for using a Visual Language Model to Provide Remote Assistance to Vehicles. 18/392,299, 2025.
- Using artificial intelligence to detect passengers in a vehicle. 18/632,989, 2025.
- System and Method for Efficient Perception in Autonomous Vehicles using a Sparse Bird's-Eye-View Backbone. pending.

Reviewer

- Neural Processing Letters (NEPL), 2019.
- European Conference on Computer Vision (ECCV), 2020.
- Conference on Computer Vision and Pattern Recognition (CVPR), 2021.
- Conference on Neural Information Processing Systems (NeurIPS), 2023.
- International Conference on Computer Vision (ICCV), 2025.

Awards

- Intel ISEF: Fourth Award in Mathematical Sciences, 2010
- University Entrance Exam: Ranked as 68 among 1.6 million examinees, 2011.

Graduate Courses

- Nonlinear Dynamical Systems, Channel and Source Coding, Matrix Analysis, Adaptation and Learning, Convex Optimization, Stochastic Processes in Dynamical Systems, Linear Dynamical Systems, Learning and Reasoning with Bayesian Networks, Machine Perception, Machine Learning Algorithms, Algorithms and Complexity, Advanced Numerical Analysis (on ODEs), Probability Theory (Measure Theory).